SIH 2026 · PS 26147 · NTRO · WINNING PLAN v3 · REVIEW-READY EDITION

Project SANJAYA

A blind signal-forensics system that **proves** its answers, says **how much more data** an open question needs, pools short captures from **different sensors**, and hands NTRO's designated sensors a **ready-to-run receiver**.

What v3 is. v2 was put through a formal design review of the kind used before a preliminary design review: every sentence read against the problem statement (PS), every citation re-checked at its source, every hardware and signal fact checked against maker or operator documentation, every number recomputed. The review raised **44 findings** (1 critical, 22 major, 21 minor; Section 1). All are dispositioned in this edition. v2 scored **6.4 / 10** on the review scorecard: a strong concept, not yet review-ready.

Verification performed for this edition (26 Sep 2026)	Result
17 research citations: title, venue, date checked on arXiv and Crossref	All exist; 1 misdescribed (arXiv 2402.17450) and 3 with imprecise date, venue or title - corrected
12 foundations (8 new: Gardner, Cluzeau, Valembois, Sicot, Marazin, Merchant, Wilson, Hanley)	All confirmed with DOIs on Crossref
Signal facts: Meteor-M LRPT, STANAG 4285, HFDL, NAVTEX, FUNcube, QO-100, ADS-B, AIS	2 errors and 3 imprecisions in v2 (FUNcube band, NAVTEX decoder; STANAG, Meteor, encryption wording) - corrected
Hardware facts: ADALM-Pluto, RTL-SDR Blog V4, KiwiSDR	Confirmed (ranges, oscillator accuracy, channel count, GPS timestamps)
Every worked number in the plan	Recomputed; formulas shown next to each number

Conventions: SNR is E_s/N_0 in the occupied bandwidth unless stated. KPIs are pre-registered **targets**, *not results. 'pw' = person-week of 40 hours. Competitor statements are what their public READMEs claim (snapshot 26 Sep 2026); code was not run.*

0. Review of v2 - scorecard and verdict

Scored against ten criteria a space-agency review board would apply. Weights reflect what decides an NTRO evaluation: correctness, coverage and innovation first; then evidence, feasibility and assurance.

Criterion	Weight	v2	Why	v3 action
Technical correctness	15%	6/10	Six factual or mathematical errors (FUNcube band, NAVTEX decoder, 'Conformal Shield' misdescribed, two-sensor clock separation underdetermined, undefined 'explained energy', STANAG wording).	Every fact re-verified; formulas and definitions added; errors corrected.
Coverage of the PS	15%	9/10	All clauses mapped; acceptance criteria loosely defined.	Requirement IDs, verification method and precise acceptance criteria (Section 4).
Innovation and uniqueness	15%	9/10	Eight crown innovations at 0 of 43 READMEs; strong narrative.	Kept; each now has honest limits, validation on real data and a TRL.
Quantitative rigour	10%	4/10	KPIs lacked observation length, SNR definition, sample sizes; 'false-certificate rate = 0' is unprovable; one ill-posed KPI.	Metric definitions, Wilson intervals, rule-of-three bound, derived worked examples.
Feasibility and schedule	10%	4/10	15 innovations plus full PS in 10 weeks with no effort model; P4 packed 7 innovations into 2 weeks.	Work-breakdown with person-weeks, capacity, 17% reserve, priority classes, gates.
Verification and validation	10%	5/10	No requirement traceability, no independence, no pre-registration.	V&V plan with independent verification and sealed test sets (Section 9).
Real-data realism	10%	7/10	Strong tiers; two wrong specifics; missed multi-site, Doppler and identity-labelled sources.	Multi-site KiwiSDR, SGP4 Doppler truth, ADS-B identities, QO-100 LDPC, SatNOGS archive.
Risk management	5%	5/10	Risks listed without likelihood, consequence, owner or trigger; no venue fallback.	5 x 5 register with owners and triggers; fallback for every demo beat.
Security and compliance	5%	3/10	Nothing on hostile input, supply chain, zero-network proof, licences, spectrum law.	Assurance section: fuzzing, SBOM, pinned offline deps, network-isolation test, legal.
Clarity and presentation	5%	8/10	Clear; some informal wording ('goated').	Formal register throughout.
Weighted total	100%	**6.4/10**	Strong concept; would not pass a design review with these findings open.	All 44 findings dispositioned (Section 1).

What changed from v2 to v3

Area	v2	v3
Evidence standard	Facts from memory	Every citation and hardware/signal fact verified at source; DOIs and arXiv IDs listed
KPIs	Single numbers	Defined metrics, conditions (SNR, data length), sample sizes, confidence intervals
Proof-of-Decode	Energy + whiteness	Four calibrated tests incl. cyclostationarity at the symbol rate; near-miss rejection; certificate lists what it could and could not rule out
Insufficient data	Claimed	Quantified: hard-decision rank methods need ~24 h of signal for a 1,500-bit interleaver (derivation in C2); SANJAYA's answer measured against that
Real data	4 tiers	Adds multi-site KiwiSDR (GPS-timestamped), SGP4 Doppler truth, ADS-B identities, QO-100 DVB-S2 (real LDPC), SatNOGS archive
Programme	Phases	Work breakdown (31.2 pw P1 vs 37.5 pw capacity), priority classes, gates G0-G5, TRL now and at finale
Assurance	Absent	Hostile-input fuzzing, SBOM, pinned offline dependencies, network-isolation proof, signing keys, licences, spectrum law

1. Findings register (review item discrepancies)

Severity: **C** critical (would sink the evaluation), **M** major (an expert judge would catch it), **m** minor. Every finding is closed in the section named in the last column.

ID	Sev	Where (v2)	Finding	Disposition in v3
R01	M	C4	'Separates transmitter from sensor clock error when two sensors hear one emitter' is underdetermined: 2 equations, 3 unknowns.	Network clock solution with one GPS/atomic reference fixing the gauge. §5 C4
R02	M	C1	'Explained energy 99.1 %' undefined. At 14.2 dB SNR the signal is only 96.3 % of in-band energy, so the figure is only valid if normalised by signal energy.	Defined $\eta = 1 - (P_r - P_n)/(P_x - P_n)$ with P_n from the Sensor Twin. §5 C1, §9
R03	M	C1 KPI	'False-certificate rate = 0' cannot be demonstrated; zero observed failures only bounds the rate.	Rule of three: 0 false PASS in ≥ 30,000 near-miss trials → P(PASS | wrong) ≤ 10^{-4} (95 %). §5 C1
R04	M	C3 KPI	'Stitched curve reaches 95 % with less than half the single-file data' is ill-posed: pooling cannot beat equal total data.	KPI = alignment loss ≤ 10 % versus one capture of equal total length. §5 C3
R05	M	Tier 3	'Amateur satellites with AO-40-style FEC (UHF 435-438 MHz)' with BPSK is wrong: FUNcube-family BPSK is VHF (145.935 MHz for AO-73); the UHF AO-40-style examples listed by gr-satellites use FSK.	Row corrected with verified frequencies. §7
R06	M	Tier 3	multimon-ng listed as a NAVTEX decoder; its mode list has no NAVTEX.	Truth source is fldigi. §7
R07	M	Refs	arXiv 2402.17450 ('Conformal Shield') is an adversarial-attack detector for AMC that uses conformal prediction, not a paper on conformal prediction sets.	Described correctly; Angelopoulos & Bates (arXiv 2107.07511) added as the conformal reference. §13
R08	M	Tier 3	'Only public, unencrypted broadcasts' contradicts STANAG 4285, whose payloads are typically encrypted.	Policy restated: modem and framing layers only; payloads never decrypted or displayed. §7
R09	m	Tier 3	STANAG 4285 given as '8PSK 2400 Bd, convolutional interleaver'. Verified: BPSK/QPSK/8PSK by data rate; rate-1/2 K=7 code; short or long interleaving (type not asserted).	Corrected. §7
R10	m	Tier 3	Meteor LRPT called 'the entire PS chain'; it has no LDPC. Also 72 kBd (80 kBd on some passes); N2-3 antenna only partly deployed.	Precise wording; LDPC covered by QO-100 DVB-S2 (stretch) and HIL. §7
R11	m	C2	Worked example '14 more codewords, 2.3 s' was not derived; 'codeword' undefined.	Derived planning model with formula and exact numbers. §5 C2
R12	m	C2	Data-need claim attributed to arXiv 2601.15903 without verifying that it states it.	Rank criterion attributed to Sicot, Houcke, Barbier (Signal Processing, 2009). §5 C2
R13	m	C5	'gr-digital / gr-fec blocks' - the GNU Radio core has no Reed-Solomon or generic de-interleaver.	Out-of-tree blocks named; bit-exact release gate. §5 C5
R14	m	C5 KPI	'Decodes 100 % of frames proven offline' mixes offline and live conditions.	Two KPIs: bit-exact on the source recording; live FER within 1 dB. §5 C5
R15	m	§6 HIL	'HF fading on real hardware' overclaims: Pluto tunes 325 MHz-3.8 GHz; fading is emulated at baseband on a UHF carrier.	Wording corrected. §7
R16	m	§0	'Five of them are new in v2' - six are new (C2-C7); C1 and C8 evolve v1 ideas.	Corrected. §5
R17	m	Refs	Non-binary LDPC paper is in the 2026 issue of Digital Communications and Networks; RS paper venue is IEEE Communications Letters; IET title abbreviated.	Full, verified citations with DOIs. §13
R18	m	§2, §5	'0 of 43' reads as a fact about competitors' code; it is a statement about README claims.	Method and snapshot date stated wherever counts appear. §3, §6
R19	m	§2	'3 show a verified real capture' - we did not verify them.	'3 describe a real capture'. §3
R20	m	C7	WAV-Lifting implied FM audio can be inverted to IQ; FM demodulation discards RF phase.	Honest limit: linear chains lifted; FM message analysed instead. §5 C7
R21	m	C6	Counter chronology ignored counter wrap-around.	Formula with modulus 2^b and elapsed-time disambiguation. §5 C6
R22	m	C3	Stitching alignment assumed; exact only with a sync marker.	Honest limit and joint phase search stated; per-capture proof ejects mis-associations. §5 C3
R23	M	§3 KPIs	KPIs lack SNR definition, observation length and sample size; identification targets at BER 10^{-2} with no data length are not credible for large interleavers.	Metric definitions and conditioned acceptance criteria. §4, §9
R24	m	§3 KPI	Fs target '0.1 %' is weak and inconsistent with the 1 ppm Sensor Twin KPI.	≤ 10 ppm for standard-anchored captures ≥ 10 s. §4
R25	C	§8	No effort model: 8 crown + 7 depth innovations + full PS in 10 weeks; 7 innovations in 2 weeks. Schedule not credible.	Work breakdown, capacity, 17 % reserve, P1/P2/P3 classes, minimum winning configuration. §10
R26	M	§8	No gates with entry/exit criteria; no rule for activating stretch work.	Gates G0-G5; P2 only from measured reserve at G3. §10
R27	M	all	No technology readiness levels; NTRO and ISRO reviewers think in TRLs.	TRL today and at finale per element (NASA NPR 7123.1 definitions). §10

ID	Sev	Where (v2)	Finding	Disposition in v3
R28	M	§10	Risks unscored; no owners or triggers.	5 x 5 risk register. §11
R29	M	§9	Live RF demo assumes a transmitter is allowed at the venue.	Every demo beat has a fallback (sealed files, throttled replay). §12
R30	m	§8	No assumptions register (dates, offline mode, single-carrier scope, finale data).	Assumptions register. §2
R31	m	§8	Per-person load not checked.	Load per role ≤ 6.25 pw. §10
R32	M	§3	No requirement IDs or verification methods.	R-01...R-15 with Test / Analysis / Inspection / Demonstration. §4
R33	M	§6	No independence: the team would grade its own work.	Independent verification with sealed test sets. §9
R34	M	§7	Hostile input not considered: NTRO ingests files from untrusted sources.	Coverage-guided fuzzing of all parsers; least-privilege engine. §8
R35	M	§7	No supply-chain control (SBOM, pinned hashes) for an air-gapped deployment.	Offline wheelhouse with pinned hashes; SBOM per release. §8
R36	M	§7	'No network at runtime' asserted, not verified.	Network-isolation test with packet capture in acceptance. §8
R37	m	§7	Licences unaddressed (several oracles are copyleft).	Oracles used only in the test harness; licence register. §8
R38	m	§6	Spectrum law and receive-side legality not addressed.	Legal and ethics box. §7
R39	m	§6	No data-management plan.	SigMF metadata, checksums, manifests for every capture. §7
R40	M	C3	Stitching demonstrated only on lab receivers, not on 'different sensors and different locations'.	Simultaneous GPS-timestamped recordings from ≥ 3 public KiwiSDRs in different countries. §7
R41	M	C4	No independent physical truth for the frequency/timing estimators.	Satellite Doppler curve versus SGP4 prediction from public TLEs. §7
R42	M	C6	Physical-trait passports validated only on the bench.	ADS-B: every aircraft self-identifies (24-bit ICAO address) - thousands of real transmitters with truth. §7
R43	M	Tier 3	No real LDPC signal in the plan.	QO-100 wideband beacon (DVB-S2, LDPC + BCH) as stretch. §7
R44	m	§0	Informal wording in a formal plan.	Formal register. all

Totals: 44 findings - 1 critical, 22 major, 21 minor. All dispositioned.

2. Mission, scope and assumptions

Mission. Turn any recording from any NTRO sensor into a **proven** transmitter recipe - sampling frequency, modulation, interleaving, FEC, scrambling, frame structure, header and payload - automatically, offline, with a statement of what the data can and cannot support, and deliver that recipe back to designated sensors as something they can run.

In scope	Out of scope (by design)
Recorded .IQ and .wav files, HF / VHF / UHF, few kHz to MHz bandwidths; single-carrier PSK / QAM / FSK / MSK families; block, convolutional, diagonal, pseudo-random interleaving; convolutional, RS, concatenated, LDPC codes; scramblers; framing; header / payload.	Decryption of payloads; live SDR control as a requirement (supported as a bonus); OFDM unless the SPOC asks for it; cloud services; LLM chat interfaces.

Minimum winning configuration (what must exist at the finale, whatever happens)

- 100 % of the PS (R-01 to R-15) on the bench and on real signals with independent truth.
- C1 Proof-of-Decode, C2 Data-Sufficiency, C3 Evidence Stitching with C4-lite, C5 Recipe Compiler - the answer to the PS's two central complaints (insufficient data, different sensors) and its output requirement (designated sensors).
- A real signal decoded end to end with independent confirmation (Meteor-M LRPT versus SatDump), and multi-site real stitching (KiwiSDR).

Assumptions register (to confirm with the SPOC / mentor)

ID	Assumption	If wrong
A-01	Analysis of recorded files is the requirement; live reception is a bonus.	C5 already produces live receivers; add a live-input path (1 pw from reserve).
A-02	Recordings may lack metadata; sampling frequency may be unknown.	None - both cases designed for.
A-03	Signals of interest are single-carrier; OFDM not required.	Add OFDM parameter estimation as P2 (1.5 pw).
A-04	Payload decryption is out of scope.	Out of scope regardless; report structure only.
A-05	Finale evaluation may use NTRO-provided files of unknown content.	Designed for; C2 reports when a file cannot support a decision.
A-06	Target hardware is a laptop class machine, no GPU required.	GPU path optional; nothing depends on it.
A-07	About ten build weeks between idea selection and finale; SIH 2026 dates to be confirmed on sih.gov.in.	Re-plan with the same WBS; P2/P3 absorb the change.
A-08	Six members at about 25 h per week each (37.5 pw).	Descoping order in §10.

3. The field: 43 public competitors

Method: the 43 ranked repositories in repos.pdf; each README searched for every capability, matching sentences read by hand; snapshot 26 Sep 2026. Counts are README claims, not verified behaviour.

Capability	READMEs claiming it	Best in field	Limitation
Loader, PSD, waterfall, constellation	~40	sanket (C++ mmap), harikesh (WebGL)	Table stakes.
PSK/QAM/FSK demodulation	~28	NaidhruvK (lock-check, equaliser)	AWGN-tuned in most.
Fixed-catalogue FEC (K=7 Viterbi, RS(255,223))	~25	Gururaghavendra123 (CRC-gated search)	Blind only inside the catalogue.
ML modulation classifier	~22	RadioFry	Trained on synthetic RadioML.
UNKNOWN / provenance / abstain labels	~20	karurravishanker	Labels uncertainty; never reduces it.
GF(2) rank interleaver ID (hard bits)	~6	ShantanuSomwanshi (size, PN seed)	Needs error-free rows: data-hungry (§5 C2).
Multi-signal detection	~6	aayushkasurde	Detects; never cancels.
Real over-the-air capture	3 describe one	ICHNOVA (KiwiSDR time stations)	No real signal with a full PHY + coding chain and independent truth.
CRC library sweep	2	yoganandasp15, aayushkasurde	Known CRCs only.
Protocol profiles	1	yoganandasp15	No sample-rate or clock back-solving.
Voice / CW / noise-only labels	~6	yoganandasp15 (NO SIGNAL verdict, CW class, voice formants)	Labels only; none uses them to decide whether to run the decode chain.

Where every competitor stops

#	Wall	Evidence (README claims)
1	One file at a time	None mentions combining captures.
2	One sensor at a time	Three restate the PS sentence; none models sensors.
3	No answer to 'is this enough data?'	None estimates data needed.
4	Output ends in a report	None generates a receiver.
5	Truth from their own generator	33 describe synthetic or generated test data.
6	Catalogue-bound coding layer	LDPC is 'pick from list' or 'not identifiable'.
7	Scramblers not identified	Fixed descramblers at most.
8	.wav = stereo IQ or Hilbert of mono	Receiver audio chains not modelled.
9	Bit-level verification at best	CRC / re-encode BER only.
10	Models never improve in use	Retrain scripts on synthetic data.

4. Pillar A - the expected solution as verifiable requirements

> **PS, expected solution (verbatim):** *"The expected system should improve feature visibility of signals with the help of GUI, enable automated signal analysis to identify spectral features such as sampling frequency, constellation plot, water fall (time-frequency domain), demodulate signals, carry out de-interleaving and error correction. The output can then be used to carry out correlation of bit stream for identification of header and payload."*

Verification method: **T** test, **A** analysis, **I** inspection, **D** demonstration (NASA Systems Engineering Handbook, SP-2016-6105 Rev 2). Reference laptop: 8-core x86-64, 16 GB RAM, integrated GPU, SSD.

ID	PS clause	Design	Verify	Acceptance criterion
R-01	.IQ or .wav input	WAV parse (mono real / stereo IQ, 8-32 bit, float); raw IQ dtype {u8,i8,i16,i32,f16,f32,f64} x endianness x I/Q order sniff with confidence; SigMF sidecar; manual override; C7 for receiver audio.	T	Correct format on 100 % of 200 bench files ≥ 64 kB at SNR ≥ 0 dB; every ambiguous case surfaced, never silent.
R-02	(i) Sampling frequency	Header / SigMF; else back-solved from a recognised standard's symbol clock and the anti-alias edge; sensor clock error reported.	T, A	≤ 10 ppm on standard-anchored captures ≥ 10 s at SNR ≥ 10 dB; otherwise 'Fs unknown' stated with normalised units.
R-03	(i) Other parameters	Centre offset, occupied bandwidth, SNR, symbol rate, roll-off, FSK tones and index, burst timing.	T	Symbol rate ≤ 10^{-4} relative (≥ 1,000 symbols, SNR ≥ 10 dB); SNR ±1 dB (SNR ≥ 0 dB); bandwidth ±5 %.
R-04	(i) Modulation	Three experts fused; calibrated probabilities; split-conformal 95 % prediction set; open-set 'novel'.	T	Top-1 ≥ 98 % at SNR ≥ 10 dB with ≥ 1,000 symbols over 11 classes (BPSK, QPSK, OQPSK, 8PSK, 16QAM, 64QAM, 2/4/8-FSK, MSK, GMSK); set coverage 0.95 ± 0.01 (n ≥ 2,000).
R-05	(i) FEC	Blind conv (rate, K, generators), RS (n, k), LDPC (catalogue by syndrome), concatenation by peeling.	T	P_{id} vs BER and data-length curves published; ≥ 95 % at BER ≤ 10^{-3} with ≤ 10^{5} coded bits for convolutional codes (K ≤ 9) and ≤ 10^{6} for RS (n ≤ 255); characterised to BER 10^{-2}.
R-06	(i) Interleaving	Block, convolutional, diagonal, pseudo-random; soft rank profile joint with code ID; PN seed library.	T	When C2 declares the data sufficient: P_{id} ≥ 95 % (periods ≤ 4,096 bits, BER ≤ 10^{-3}); data required vs period and BER reported as curves; characterised to BER 10^{-2}.
R-07	(ii) Demodulate FSK, QAM, PSK	RRC matched filter, Oerder-Meyr + Gardner timing, M-th power + Costas carrier, CMA → DD-LMS equaliser, FSK bank; soft LLRs.	T	BER within 0.5 dB of theory at BER 10^{-4} (AWGN) per scheme; HF (Watterson) curves reported.
R-08	(iii) De-interleave x 4 types	Invertible operators parameterised by R-06; manual override.	T	Bit-exact round trip on 10^{4} random vectors per type.
R-09	(iv) FEC decode x 4 families	C++ soft Viterbi (K 3-9, punctured), RS Berlekamp-Massey, concatenated, LDPC normalised min-sum.	T	Within 0.2 dB at FER 10^{-3} of reference implementations: liquid-dsp or GNU Radio gr-fec (convolutional, RS), AFF3CT (LDPC).
R-10	(v) Bit-stream correlation, header and payload	Autocorrelation frame length; soft sync discovery; aligned raster; counters, length fields; CRC solved by GF(2) algebra.	T	Frame length and sync on 100 % of framed files with ≥ 20 frames at BER ≤ 10^{-3}; field class accuracy ≥ 95 % per column.
R-11	GUI and feature visibility	Built React + three.js front-end (desktop via Tauri); 5-step flow; waterfall, PSD, constellation on first view.	D, I	All three spectral features visible on first load; any clause ≤ 2 clicks; 100 MB file interactive < 2 s.
R-12	Automated	Autopilot from raw file to header / payload; Expert overrides any stage.	D, T	One click; full chain < 10 s for 10^{6} samples.
R-13	GNU Radio, Python, C++	Python orchestration; C++ kernels; GNU Radio for C5 output and as oracle.	I	Each used for a stated reason.
R-14	Output for designated sensors	Certificate, SigMF, bits, CSV, recipe file, compiled receiver (C5).	T, D	Compiled receiver bit-exact on source recording (§5 C5).
R-15	Signal type triage (first step of (i))	Before decoding, classify the file's content as digital data, analog voice, carrier / Morse or noise from spectral features (occupied band, symbol-rate line, frame-to-frame level and spectral change, single-line energy); only digital data enters the decode chain; mixed captures split by the Scene step.	T	Correct class on ≥ 95 % of 400 labelled files (synthetic + real: KiwiSDR voice and data, SatNOGS audio); verdict < 1 s per file; files too short for a verdict say so.

5. Pillar B - the USPs

Crown innovations answer sentences in the PS and are claimed by none of the 43 competitor READMEs. Six are new in this plan (C2-C7); C1 and C8 evolve v1 ideas. Each card states honest limits, real-data validation, priority class (P1 = minimum winning configuration, P2 = from reserve, P3 = bench prototype and roadmap) and technology readiness (TRL, NASA NPR 7123.1).

C1 · Proof-of-Decode and the Decode Certificate	P1 · TRL 2 → 5 · 1.5 pw
PS	*"This limitation reduces the accuracy and confidence of interpretation and data analysis that require detailed information."*
Design	Re-encode the decoded bits through the recovered chain (code → interleaver → scrambler → mapper → pulse shape), apply the estimated channel (gain, CFO, phase trajectory, timing, equaliser taps), subtract, and test the residual r. A PASS needs all four tests, with thresholds calibrated on Tiers 1-2: **(T1)** explained signal energy $\eta = 1 - (P_r - P_n)/(P_x - P_n) \geq \eta_{min}$, with noise power P_n from the Sensor Twin (C4); **(T2)** residual-to-noise ratio P_r/P_n within ±0.2 dB; **(T3)** no cyclostationary feature at the symbol rate left in r (cyclic coherence at $\alpha = 1/T$ below threshold); **(T4)** pre-whitened residual white (Ljung-Box) and Gaussian (kurtosis). CRC pass rate and syndrome weight are recorded as independent bit-level corroboration.
Certificate	SHA-256 of the input, SANJAYA version and configuration hash, full recipe, every test statistic and threshold, and the **near-miss alternatives** tested (e.g. 8PSK for QPSK, wrong roll-off, wrong interleaver period) with whether each was rejected. Ed25519 signature with an offline per-site key. Deterministic: same version + file + configuration → identical recipe.
Honest limits	A PASS means the recipe explains the recording to the noise floor. Recipes that produce identical waveforms (for example a scrambler folded into random payload) cannot be separated by the proof; structure (frames, CRC) separates them, and the certificate reports such equivalences. At low SNR near-miss rejection weakens; the certificate lists which alternatives it could not rule out.
New because	No competitor README claims waveform-level verification; the best claim bit-level checks (Gururaghavendra123, Venkata-Manoj, anshumanarchit).
Built on	Analysis-by-synthesis / successive cancellation; Gardner (IEEE SP Magazine, 1991) cyclostationarity; Ljung and Box (Biometrika, 1978); Hanley and Lippman-Hand (JAMA, 1983) rule of three.
Real validation	PASS certificates on Meteor-M LRPT and HFDL recordings must agree with SatDump and dumphfdl decodes.
KPI	P(PASS | wrong recipe) ≤ 10^{-4} (95 % upper bound: 0 false PASS in ≥ 30,000 near-miss trials); P(PASS | right recipe) ≥ 95 % at SNR ≥ 6 dB with ≥ 10^4 symbols.

C2 · Data-Sufficiency Engine and collection tasking	P1 · TRL 2 → 4 · 1.0 pw
PS	*"This data is often insufficient for fine grain analysis for parameter extraction such as modulation type, sampling rate, FEC, interleaving, etc."*
Design	Each open decision is a hypothesis set with log-evidence from calibrated likelihoods (demodulation, rank / syndrome statistics, C1 re-synthesis). Decide when the best posterior ≥ $1 - \alpha$ ($\alpha = 10^{-3}$). Otherwise measure the evidence growth rate D (nats per second, the slope of the log Bayes factor between the top two hypotheses) and predict $\Delta T \approx [\ln((1-\alpha)/\alpha) - \ln B] / D$ (Wald's sequential-test approximation). The result becomes a **tasking note**: centre frequency, occupied bandwidth, minimum complex sample rate (≥ 1.25 x occupied bandwidth), duration (ΔT x 1.3), minimum SNR, and the best sensor from the Twin.
Why it matters	For methods that need error-free rows (the hard-decision rank criterion; Sicot, Houcke, Barbier, Signal Processing 2009), rows needed ≈ $2N / (1-p)^N$ for period N bits at bit error rate p. At p = 3 x 10^{-3} and 4.8 kbit/s: **N = 96 → 256 rows, 24.6 kbit, 5.1 s**; **N = 1,500 → 2.7 x 10^5 rows, 4.1 x 10^8 bits, about 23.6 hours**. This is why 'insufficient data' is real, why soft-decision and code-aided identification (D1) and stitching (C3) matter, and why an analyst needs to be told the number.
Honest limits	Predictions assume the emitter keeps one mode; a mode change shows up as a collapsing evidence rate and is flagged, not extrapolated.
New because	No competitor README estimates the data a decision needs; they print a result or UNKNOWN.
Built on	Wald (Annals of Mathematical Statistics, 1945); Bayesian model comparison; Sicot et al. (2009) rank criterion.
KPI	Median $\lvert\Delta T_{pred} - \Delta T_{actual}\rvert / \Delta T_{actual} \leq 30$ % (90th percentile ≤ 60 %) on Tiers 1-2; reported on Tier 3.

C3 · Evidence Stitching across captures, sensors and locations	P1 · TRL 2 → 5 · 1.5 pw
PS	*"Since the data point are recorded from different sensors and different locations, the parameters may vary. Therefore, the data available for analysis is often insufficient..."*
Design	(1) **Associate** captures to one emitter only when recipes agree and passport distance is below threshold. (2) **Normalise** each capture with its sensor's Twin (resample to a common symbol grid, remove CFO, match gain). (3) **Align** exactly on a sync word or frame marker; otherwise search the phase per capture over the candidate period. (4) **Pool**: a code's dual space is identical in every capture, so aligned soft codeword rows are stacked into one matrix and log-likelihoods add. (5) **Safety**: C1 then runs per capture - a wrongly associated capture fails its own proof and is ejected.
Honest limits	Without a sync marker the phase search grows with the period. Captures in different emitter modes are never pooled.
New because	No competitor README combines captures; all are single-file.
Real validation	One HF transmission recorded at the same time by ≥ 3 public KiwiSDRs in different countries, with GPS-timestamped IQ - literally 'different sensors and different locations'.
KPI	Alignment loss ≤ 10 %: K captures of total length T reach within 10 % (relative) of the identification probability of one capture of length T. Association false-merge ≤ 1 %.

C4 · Sensor Twin and network clock solution	P1-lite / P2-full · TRL 2 → 5 (lite) · 0.5 + 0.5 pw
PS	*"The analysis of signals is primarily dependent on the basic characteristics of data points selected during recording of these signals."*
Design	Per sensor: sample-clock error ε_s, local-oscillator error, anti-alias response, noise-floor shape, IQ gain and phase imbalance, DC, clipping level, phase noise. Learned passively from signals of opportunity: standard-frequency broadcasts (atomic-referenced carriers), recognised standards (exact symbol clocks), empty bands, strong carriers. **Network clock solution**: each observation of emitter e at sensor s gives $\delta_{es} = \varepsilon_e + \varepsilon_s$ (after removing Doppler: SGP4 from public TLEs for satellites; negligible at ppm level for HF sky-wave). One reference - a GPS-disciplined receiver such as a KiwiSDR, or an atomic-referenced broadcast - fixes the gauge; weighted least squares gives every sensor and emitter clock with its uncertainty.
Honest limits	With no reference, sensor and emitter clocks cannot be separated (the network has one free offset); SANJAYA then reports relative clocks only.
Uses	C3 normalisation; C6 emitter clocks; metadata-lie detection (declared sample rate versus anti-alias edge and symbol clocks); Twin-conditioned training data (P2).
New because	Competitor READMEs correct DC and IQ imbalance per file; none keeps a persistent sensor model.
KPI	Sensor clock error within 1 ppm (1σ) of a GPS-disciplined reference; declared-vs-measured sample-rate discrepancies ≥ 0.5 % detected in ≥ 99 % of recordings containing a recognised standard.

C5 · Recipe Compiler: from a recording to a deployable receiver	P1 · TRL 2 → 5 · 1.5 pw
PS	*"...the resultant data is then utilised for processing signals in the designated sensors."*
Design	Input: a **certified** recipe only (C1 PASS). Output (a): a GNU Radio 3.10 flowgraph using gr-digital (symbol sync, Costas loop, constellation decoder, additive scrambler, access-code correlator) and gr-fec (convolutional decoding), plus SANJAYA out-of-tree blocks where the core has none (Reed-Solomon, generic de-interleaver, CRC-n, schema-driven frame parser). (b): a versioned recipe file (JSON, SigMF-compatible namespace) for NTRO's own receivers. (c): a recipe-matched detector - coherent gain ≈ $10 \log_{10} L$ dB for L known symbols (≈ 15 dB for a 32-symbol sync word) plus incoherent integration across frames.
Release gate	The compiled receiver, run on the source recording, must reproduce SANJAYA's decoded bits exactly before it can be exported.
Honest limits	Chains with no GNU Radio or SANJAYA block (rare) export the recipe file only.
New because	No competitor README generates a receiver; outputs end in reports or JSON.
KPI	100 % bit-exact on source recordings; live frame-error-rate curve on the hardware-in-the-loop rig within 1 dB of the offline decoder's.

C6 · Emitter Passport and Counter Chronology	P2 · TRL 2 → 4 (5 for physical traits on ADS-B) · 1.0 pw
PS	*"This creates a need for advanced data processing to extract the observation data."*
Design	Passport = recipe (exact fields) + physical traits (emitter symbol-clock error after sensor and Doppler correction, carrier offset, IQ image, phase-noise profile, turn-on transient) + protocol state (counters, sequence numbers). Recipe must match exactly; traits by Mahalanobis distance; open-set threshold → 'new emitter'. **Counter chronology**: for a b-bit counter and frame period T_f, frames between captures $n = ((c_2 - c_1) \bmod 2^b) + k \cdot 2^b$, $k = \text{round}[(\Delta t/T_f - ((c_2 - c_1) \bmod 2^b)) / 2^b]$; duty cycle = $n \cdot T_f/\Delta t$; equal counters at two sensors give a common epoch.
Honest limits	Physical traits drift with temperature and ageing; passports carry uncertainty and are re-anchored by recipe and counters.
Real validation	ADS-B at 1090 MHz: every aircraft broadcasts its 24-bit ICAO address - thousands of physical transmitters with self-declared identity, receivable across India with an RTL-SDR. (ADS-B is pulse-position modulation: used only to validate physical traits.)
KPI	Closed-set re-identification ≥ 90 % top-1 among ≥ 50 emitters; open-set AUROC ≥ 0.95; chronology exact on the rig's counters.

C7 · WAV-Lifting: undoing the receiver's audio chain	P3 · TRL 2 → 4 · 1.0 pw
PS	*"The data saved as .IQ and .wav have different parameters ... The spectral relationship from training data containing both .IQ and .wav formats can be utilised..."*
Design	Classify the .wav: stereo IQ, real IF, or receiver audio. For linear receivers (SSB / AM / CW audio) estimate sideband orientation, audio offset, passband edges and AGC trajectory, then rebuild complex baseband (analytic signal + shift), keeping sideband and spectral inversion as hypotheses.
Honest limits	FM demodulation discards RF phase: for FM audio SANJAYA analyses the demodulated message (e.g. AFSK / FSK data) and marks IQ-only features unavailable.
Real data	KiwiSDR: two of its four channels on one frequency, one GPS-timestamped IQ and one USB audio, recorded simultaneously. SatNOGS archive: demodulated audio plus decoded frames (open API, CC BY-SA); lossy audio treated as a known impairment.
KPI	Recipe agreement IQ vs lifted WAV ≥ 95 % on paired KiwiSDR recordings (linear chains).

C8 · Proof-gated Learning Flywheel (on-premises)	P3 · TRL 2 → 4 · 1.0 pw
PS	*"The spectral relationship from training data ... can be utilised for identifying signal parameters and carry out deeper analysis."*
Design	Only C1 PASS results with margin become labels. Scheduled fine-tuning of the small models; split-conformal recalibration on a fresh calibration split; prototype and passport library growth.
Safeguards	Frozen golden set never trained on; automatic rollback if golden accuracy drops by more than 1 point; per-class quotas against drift toward easy signals; active-learning queue for NOVEL and FAIL cases; model files inherit the data's classification; no network.
Honest limits	Self-training only reinforces what the system can already prove; genuinely new signal types still need an analyst (few-shot teach).
Built on	Wireless foundation models and few-shot adaptation: IQFM (arXiv 2506.06718), EMind (2508.18785), Radio-FM (2608.05793), RFPrompt (2605.03279); Angelopoulos and Bates (arXiv 2107.07511).
KPI	Golden-set accuracy non-decreasing over ≥ 3 rounds; admitted-label error bounded by the C1 false-PASS bound.

Depth USPs - the strongest coding layer in the field

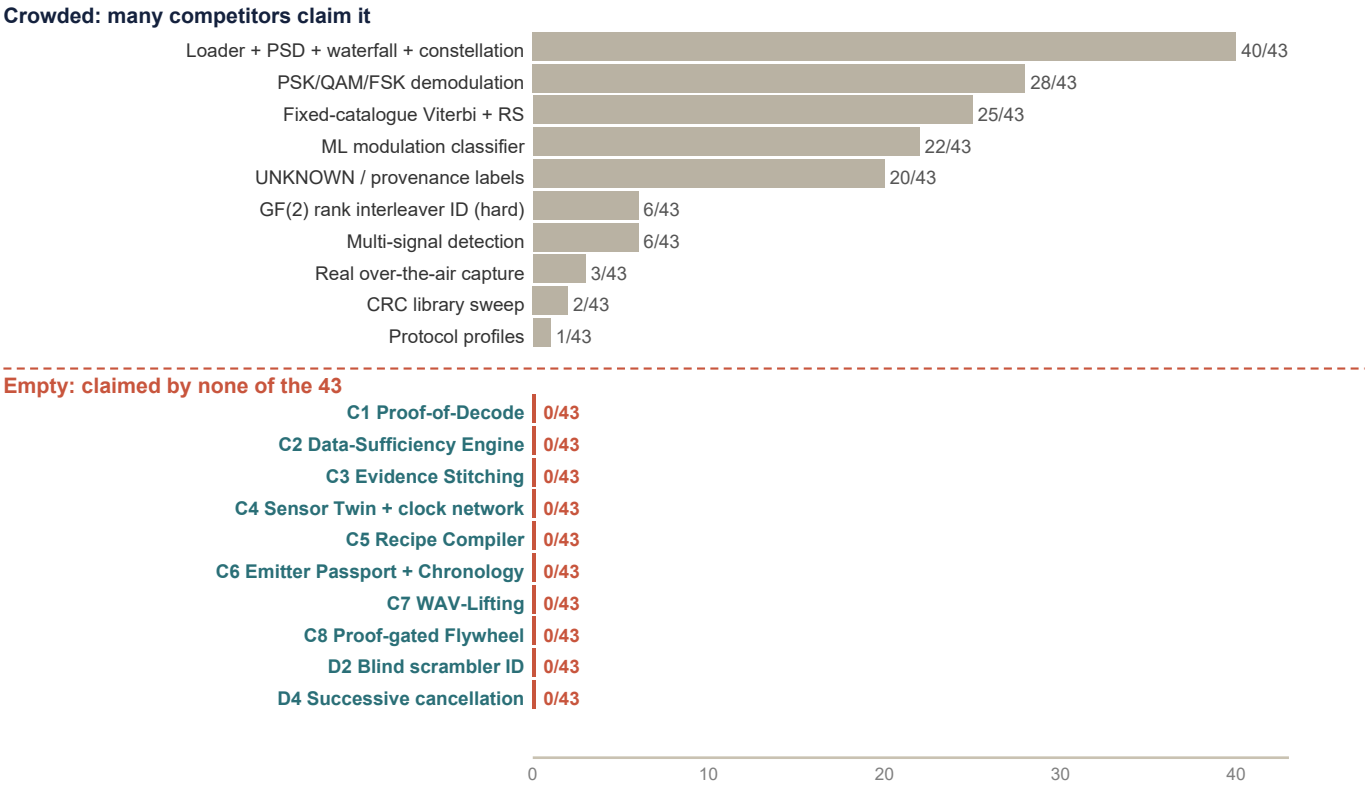
USP	What	Class	Basis
D1 Soft joint coding engine	Interleaver + code jointly from soft LLRs; rank-profile plot; peeling of concatenated codes; open-set LDPC for n ≤ 1,024 (P3).	P1 (open-set P3)	Valembois 2001; Marazin, Gautier, Burel 2011; Sicot et al. 2009; arXiv 1501.03715; arXiv 2601.15903; DCN 2026 (LDPC)
D2 Blind scrambler recovery	Additive and self-synchronising LFSRs; Berlekamp-Massey on idle runs; standard library.	P1	Cluzeau (IEEE Trans. Computers, 2007); Massey 1969; IET Comms 2026
D3 Bit-Lens with CRC solving	Aligned raster, entropy map, counters, length fields; any CRC polynomial solved in GF(2); schema export.	P1	Sensors 2024 (frame sync); Emp. SE 2026 (format inference)
D4 Scene + successive cancellation	OS-CFAR scene; proven emitters subtracted; buried emitters revealed and decoded.	P2	SIC theory; CSRD2025; ZoomSpec

USP	What	Class	Basis
D5 Signal Genome + back-solving	~30 named standards; true sample rate and sensor clock from a matched standard.	P2	Public standards
D6 HF robustness	Watterson channel (ITU-R F.1487) emulated in software and on the rig; fractionally-spaced equaliser; real HF via KiwiSDR.	P2	Watterson, Juroshek, Bensema 1970
D7 Trust layer	Calibrated probabilities, split-conformal sets (P1); open-set and few-shot teach (P2).	P1 / P2	Angelopoulos & Bates 2021; arXiv 2605.26600

Deliberately not doing: LLM chat interfaces (3 of 43 READMEs have one; a liability for an air-gapped user), cloud and login, black-box-only deep learning.

6. The High-Innovation Gap - where SANJAYA wins

Bars: number of the 43 competitor READMEs claiming each capability (keyword search + manual reading, snapshot 26 Sep 2026).



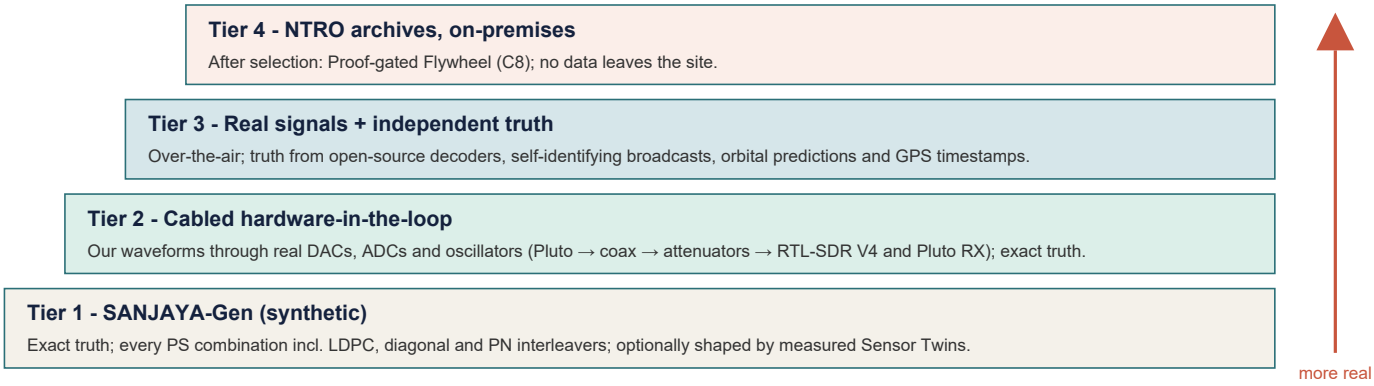
Head-to-head with the three strongest competitors (repos.pdf ranking; README claims)

Capability	ShantanuSomwanshi (9.0)	NaidhruvK (8.8)	Gururaghavendra123 (8.6)	SANJAYA (planned)
PS (i)-(v) full chain	Yes	Yes	Yes	Yes
GUI	Yes	Not evident	Yes	Yes (built)
Blind interleaver size	Yes (hard rank)	Yes (hard rank)	Catalogue	Yes (soft, joint)
Waveform-level proof	No	No	No (CRC gate)	Yes
Data needed for a decision	No	No	No	Yes
Pools captures / sensors	No	No	No	Yes
Receiver for sensors	No	No	No	Yes
Real signal, full chain, independent truth	No	No	No	Yes
Blind scrambler ID	No	Fixed descrambler	No	Yes
CRC solved (not looked up)	No	No	Known CRC check	Yes

Why this gap is durable. The crown innovations depend on each other: the Flywheel needs Proof; Stitching needs the Twin and the Data-Sufficiency maths; the Passport needs Bit-Lens counters and the clock network; the Compiler needs certified recipes. Copying one idea from a deck does not reproduce the system.

7. Beating insufficient data with real recordings

Inside the algorithm: soft decisions, joint identification and sequential stopping need less data (C2, D1); stitching and the Twin let data be pooled (C3, C4). **Outside**: a real-data engine where every recording has truth we did not create.



Tier 3 catalogue (facts verified against operator, maker and project documentation)

Signal	Verified parameters	Exercises	Independent truth	How
Meteor-M N2-4 (N2-3 weaker: antenna partly deployed)	137.9 MHz (137.1 backup); QPSK 72 kBd (80 kBd on some passes); rate-1/2 convolutional code, interleaving, Reed-Solomon over CCSDS frames	Demod, Viterbi, RS, concatenation, interleaving, framing (not LDPC)	SatDump decode	RTL-SDR V4 + V-dipole, passes over India
HFDL	1800 Bd; BPSK 300 / QPSK 600 / 8PSK 1200-1800 bit/s; FEC and interleaving per ARINC 635	Demod x 3 PSK orders, FEC, interleaving, CRC frames	dumphfdl (reads recorded CF32/CS16 IQ)	KiwiSDR IQ
STANAG 4285	1800 Hz tone, 2400 Bd; BPSK/QPSK/8PSK by rate (75-3600 bit/s); rate-1/2 K=7 code; short or long interleaving; 80-symbol preamble, 4 x 32 data + 3 x 16 probes	HF fading, equaliser, FEC, interleaving	Preamble and probe sequences from the public standard + C1 proof (payloads usually encrypted: modem layers only)	KiwiSDR IQ
NAVTEX / SITOR-B	518 kHz; 100 Bd FSK, 170 Hz shift; time-diversity error correction	FSK, framing	fldigi	KiwiSDR
AO-40-FEC satellites	FUNcube family: BPSK 1200 Bd on VHF (AO-73: 145.935 MHz); some UHF satellites use AO-40-style FEC on FSK near 437 MHz	Concatenated RS + convolutional, interleaving, scrambling	gr-satellites	RTL-SDR; check live status on SatNOGS
QO-100 WB beacon (stretch: real LDPC)	10,491.5 MHz; DVB-S2, QPSK 1.5 MS/s, FEC 4/5 (LDPC + BCH); India within footprint	LDPC, BCH, scrambling, framing	Published beacon parameters; transport-stream CRC and continuity checks	60-90 cm dish + PLL LNB → 741.5 MHz IF → Pluto
ADS-B (C6 only)	1090 MHz; pulse-position modulation; 24-bit ICAO address in every message	Physical-trait passports	Aircraft's own ICAO address (dump1090)	RTL-SDR anywhere near air routes
AIS (optional)	161.975 / 162.025 MHz; GMSK 9600 bit/s; HDLC, NRZI, bit stuffing, CRC-16	GMSK, framing; MMSI identities	AIS-catcher	Coastal sites only

Three sources of truth that need no decoder

Source	What it proves	Used by
Multi-site KiwiSDR: up to 4 channels per receiver; IQ saved with GPS timestamps (kiwirecorder 'kiwi-wav'); receivers across many countries	Same transmission from different sensors and locations, time-aligned to GPS - the PS's own scenario	C3, C4
SGP4 Doppler: public TLEs predict each satellite pass's Doppler curve	Independent physical truth for our carrier and symbol-rate estimators; residual = transmitter + sensor clock	C4, R-02, R-03
Atomic-referenced broadcasts (standard-frequency and time stations on HF)	Absolute reference for sensor clock error	C4
SatNOGS archive: demodulated audio, waterfalls and decoded frames (open API, CC BY-SA)	Receiver-audio recordings with decoded truth, at scale	C7, C8

Tier 2 rig (verified specifications)

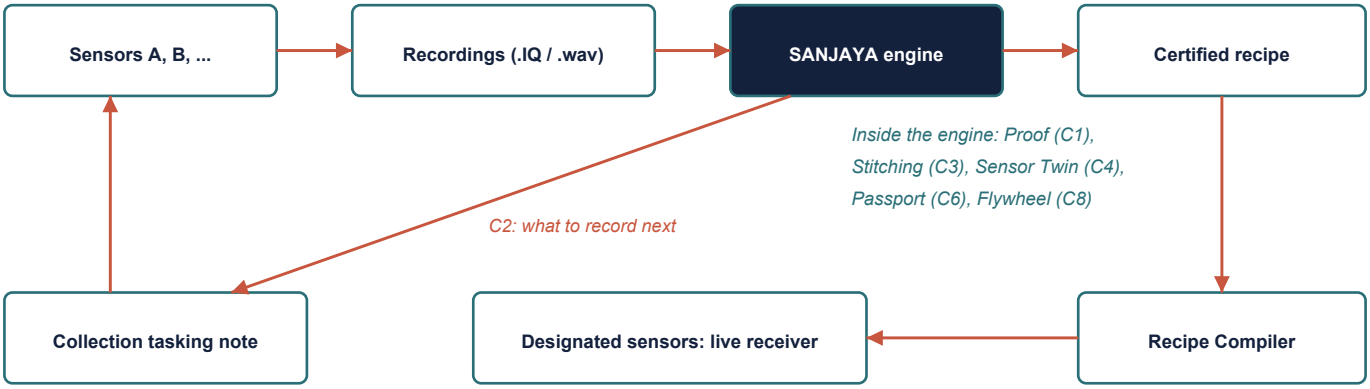
Item	Specification	Role	Approx. cost
ADALM-Pluto	325 MHz-3.8 GHz; one TX, one RX; reference ±25 ppm	Transmitter; also a 'poor-clock' receiver	₹20-28k

Item	Specification	Role	Approx. cost
2 x RTL-SDR Blog V4	500 kHz-1.766 GHz; 1 ppm TCXO; HF via built-in upconverter	'Good-clock' sensors; Meteor, ADS-B	₹8-10k
Attenuators (≥ 50 dB total), splitter, coax, shielding	Conducted path only	Two sensors fed simultaneously	₹3-5k
V-dipole (137 MHz); optional dish + PLL LNB	Tier 3 reception	Meteor; QO-100 (stretch)	₹1-2k (+ ₹5-8k)

Legal and ethics. Transmit: conducted only - coax into ≥ 50 dB of attenuation, no antenna, shielded. Where a test signal fits it, the licence-exempt 433.05-434.79 MHz band (WPC G.S.R. 347(E), 2022: ≤ 10 mW e.r.p., ≤ 10 kHz channel bandwidth, ≤ 10 % duty cycle) is used as a second safeguard; wider test signals stay strictly conducted. **Receive**: public broadcasts only; confirm equipment and reception rules (Indian Wireless Telegraphy Act, 1933; WPC) with the institute. **Content**: payloads never decrypted or displayed; identities (ICAO, MMSI) used only as truth labels, never published. **KiwiSDR**: respect each operator's time limits and credit them.

Data management. Every capture stored as SigMF (data + metadata: UTC time, receiver, location, antenna, gain, sample rate, centre frequency, truth source), with SHA-256 checksums in a versioned manifest; synthetic sets regenerated from recorded seeds.

8. System, interfaces and assurance



Only certified recipes compile (C5 gate); C2 closes the loop back to collection.

Layer	Technology	Interface
Engine	Python 3.11 + NumPy/SciPy; C++17 (pybind11): Viterbi, LDPC min-sum, bit-packed GF(2) elimination; *galois* for GF(2^m)	Versioned local API (localhost only)
GUI	React + three.js front-end (built); Tauri desktop shell	Consumes the local API; no remote origins
Recipe	JSON schema, semantic-versioned; SigMF-compatible namespace	Shared by certificate, compiler and sensors
Oracles	AFF3CT, liquid-dsp, GNU Radio blocks; Tier-3 decoders (SatDump, gr-satellites, dumphfdl, fldigi, dump1090, AIS-catcher)	Test harness only - not shipped
ML	PyTorch small encoders (< 5 M parameters, CPU); split conformal	Model files versioned with data manifests

Assurance for an intelligence user

Concern	Control	Verified by
Hostile input (files from untrusted sources)	Coverage-guided fuzzing of WAV, raw-IQ and SigMF parsers; strict size and type limits; engine runs least-privilege, isolated from the UI	≥ 24 CPU-hours fuzzing, zero crashes or hangs (T)
Network leakage	No network code paths at runtime; local API bound to 127.0.0.1	Full demo in a network-isolated sandbox with packet capture: zero outbound packets (T)
Supply chain	Offline wheelhouse; every dependency pinned by hash; SBOM (CycloneDX) per release; signed installer	Build from wheelhouse on a clean offline machine (D)
Integrity of certificates	Ed25519 key generated offline per site; private key never exported; configuration and version hashed into every certificate	Reproduce a certificate on a second machine (T)
Licences	Licence register; copyleft tools used only as oracles in the harness; SANJAYA's own GNU Radio blocks released with source	Inspection of release bundle (I)

9. Verification and validation plan

Level	What	Pass rule
Unit	Every estimator, decoder and parser, checked against oracles (AFF3CT, liquid-dsp, GNU Radio) and analytic cases	Oracle agreement within stated tolerance
Integration	Full chain on Tier 1 with known truth	Recipe exact; decoded bits exact
System	Tiers 1-3, including near-miss wrong hypotheses for C1	Section 4 acceptance criteria
Independent	A member who did not build the module runs sealed test sets (Tier 1 seeds and Tier 2-3 captures never shown to developers)	Numbers reported as measured, with intervals
Operational	Demo script on the frozen build, five rehearsals, one with an injected failure	Under 7 minutes; fallbacks exercised

Metric definitions (pre-registered: frozen in this document before any measurement)

Metric	Definition
SNR	E_s/N_0 in the occupied bandwidth, measured by the M2M4 estimator and cross-checked against the PSD.
P_{id}	Fraction of trials in which the full parameter is right (e.g. interleaver type and period). Reported with a 95 % Wilson interval; n ≥ 500 per condition (half-width ≤ 1.9 % at 0.95).
BER	Always labelled pre-FEC (after demodulation) or post-FEC.
False PASS	P(PASS | recipe wrong), over deliberately wrong near-miss recipes; when zero are observed, bounded by the rule of three (3/n).
η, ρ	Explained signal energy and residual-to-noise ratio as defined in C1.
Alignment loss	$1 - P_{id}$(K stitched captures, total T) / P_{id}(one capture, length T).
ΔT error	$\lvert\Delta T_{pred} - \Delta T_{actual}\rvert / \Delta T_{actual}$, where ΔT_{actual} is the extra duration after which the decision actually became decisive.

Every run records code version, configuration hash, dataset manifest hash and seeds, so any number in the pitch can be reproduced on demand. A fast regression suite runs on every merge; the full bench runs nightly.

10. Programme: effort, capacity, gates, readiness

Capacity: 6 members x 25 h/week x 10 weeks = 1,500 h = **37.5 pw**. P1 is planned at 31.2 pw, leaving **6.3 pw (17 %) reserve**. P2 draws on reserve only after gate G3; P3 is bench prototypes and roadmap.

WP	Content	Class	pw	Owner
WP01	Ingest and forensics (R-01)	P1	1.0	DSP
WP02	Measurements, Fs back-solve (R-02, R-03)	P1	1.5	DSP
WP03	Receivers PSK/QAM/FSK/GMSK + equaliser (R-07)	P1	3.0	DSP
WP04	AMC experts, calibration, conformal (R-04, D7)	P1	1.5	ML
WP05	FEC decoders in C++ (R-09)	P1	2.5	Coding
WP06	Blind coding layer, hard → soft (R-05, R-06, D1)	P1	3.5	Coding
WP07	Scrambler recovery (D2)	P1	1.0	Protocol
WP08	Bit-Lens + CRC solving (R-10, D3)	P1	1.5	Protocol
WP09	C1 Proof-of-Decode + certificate	P1	1.5	ML
WP10	C2 Data-Sufficiency + tasking	P1	1.0	ML
WP11	C4-lite Twin + C3 Stitching	P1	2.0	Data
WP12	C5 Recipe Compiler + OOT blocks	P1	1.5	Product
WP13	SANJAYA-Gen, bench, CI	P1	2.0	Protocol
WP14	Real-data engine: rig, Meteor, multi-site KiwiSDR, HFDL, truth pipelines	P1	2.0	Data
WP15	UI-engine integration (R-11, R-12)	P1	1.5	Product
WP16	Assurance: fuzzing, isolation test, SBOM, installer	P1	1.5	Product
WP17	Independent verification runs + KPI report	P1	1.0	Protocol*
WP18	Pitch, demo, rehearsals	P1	1.0	Data (lead)
WP19	Signal type triage: digital / voice / CW / noise gate (R-15)	P1	0.7	ML
	P1 total		**31.2**	reserve 6.3 pw
P2	C4-full, C6 Passport + chronology, D4 cancellation, D5 Genome, D6 real-HF tuning, D7 open-set	P2	4.0	from reserve after G3
P3	C7 WAV-Lifting, C8 Flywheel, open-set LDPC, QO-100 real LDPC	P3	4.5	bench / roadmap

** The protocol lead coordinates independent verification but never verifies WP07, WP08 or WP13, which they build. Load per role (P1): DSP 5.5, Coding 6.0, Protocol 5.5, ML 4.7, Product 4.5, Data lead 5.0 pw - all within 6.25 pw each; Coding is the tightest.*

Gates

Gate	When	Exit criterion
G0	Idea submission	Deck from this plan; UI clip; C1 clip on a synthetic file
G1	End W2	Spine decodes Tier 1 and Tier 2 files in the UI; bench and CI running
G2	End W4	Blind coding core + C1 on Tiers 1-2; first Meteor decode matches SatDump
G3	End W6	R-01 to R-15 pass on the bench; **decide P2 activation from measured reserve**
G4	End W8	C2, C3 (multi-site real), C5 (bit-exact) complete; demo script frozen
G5	End W9	Independent KPI report; feature freeze; W10 rehearsal only

Technology readiness (NASA NPR 7123.1: TRL 2 concept formulated; 4 validated in the lab; 5 validated in a relevant environment)

Element	Today	Finale target	Evidence at target
C1 Proof	2	5	Certificates on real Meteor and HFDL recordings agree with independent decoders
C2 Data-Sufficiency	2	4	Prediction error on Tiers 1-2
C3 Stitching + C4-lite	2	5	Multi-site KiwiSDR stitching; sensor clock versus GPS reference
C5 Compiler	2	5	Bit-exact on a real recording; live on the rig
C6 Passport	2	4 (5 traits)	Rig counters; ADS-B identities

Element	Today	Finale target	Evidence at target
C7, C8	2	4	Bench prototypes
Front-end	4	5	Built today with demo data; wired to the engine on real recordings

11. Risk register

Likelihood (L) and consequence (C) from 1 to 5; score = L x C. Red ≥ 15, amber 8-14.

ID	Risk	L	C	Score	Mitigation	Trigger	Owner
K01	Real signals need more data than captures hold (large interleavers)	4	4	16	Sync-aided stitching (C3); soft methods (D1); C2 reports honestly	P_{id} on Tier 3 < 80 % at G3	Coding
K02	Schedule overrun	4	4	16	WBS with 17 % reserve; gates; P2/P3 descoping order	Any WP > 25 % over at a gate	Lead
K03	Demo failure at the venue	3	4	12	Frozen build; 5 rehearsals; backup laptop; fallback per beat (§12)	Any rehearsal failure	Product
K04	False certificates	2	5	10	Four-test proof; calibrated thresholds; near-miss testing; independent runs	Any false PASS in bench	ML
K05	RF equipment not allowed at the venue	3	3	9	Cabled, low power; file-replay fallback	Venue rules	Lead
K06	Meteor reception poor or satellite status changes	3	3	9	Prefer N2-4; several passes per day; HFDL and STANAG via KiwiSDR	No clean pass by W2	Data
K07	Coding lead overloaded (6.0 of 6.25 pw)	3	3	9	Pair with ML lead on WP06; move open-set LDPC to P3	WP06 late at G2	Lead
K08	Legal or institute approval for the rig	2	4	8	Conducted only; licence-exempt band where applicable; early approval	No approval by W1	Data
K09	Hostile or malformed files crash the engine	2	4	8	Fuzzing; limits; isolation	Any crash in fuzzing	Product
K10	Team availability (examinations)	3	3	9	Reserve; cross-trained pairs	> 1 member away > 1 week	Lead
K11	Stitching associates the wrong captures	2	3	6	Recipe + passport association; per-capture proof ejects	False-merge > 1 %	Data
K12	Dependency or licence problem	2	3	6	Pinned wheelhouse; licence register	Build fails offline	Product

12. Demonstration with a fallback for every beat

Time	Beat	What judges see	Fallback
0:00	The problem	A real Meteor-M recording: a noisy waterfall.	Recorded screen video
0:30	Autopilot on real data	Every layer recovered; matches SatDump.	Cached results, same UI
1:40	Proof	Ridge sinks into the noise; certificate prints; a near-miss (8PSK) is rejected on screen.	Offline; no hardware needed
2:30	Insufficient data	0.4 s capture → tasking note; multi-site KiwiSDR captures stitched → proven.	Offline; deterministic
3:30	Mystery signal	A judge chooses modulation, interleaver, code, scrambler; the rig transmits over cable; SANJAYA recovers all.	Judge picks one of 20 sealed pre-generated files
4:40	Recipe Compiler	One click: a GNU Radio receiver decodes the live cable stream.	Throttled file-source replay
5:40	Passport + hidden signal	Only if P2 delivered; otherwise skipped (demo ends by 6:00).	Skip
6:30	Numbers	Independent KPI table with confidence intervals; close on the mission sentence.	Printed handout

Questions to answer cold

Question	Answer
How do you know it is right with no ground truth?	Four-test waveform proof with near-miss rejection; agreement with independent decoders on real signals; bounded false-PASS rate.
How much signal do you need?	It depends on the question; SANJAYA computes it (C2). Example: hard-rank methods need about 23.6 h for a 1,500-bit interleaver at BER 3 x 10^{-3}; we show our measured curve against that.
Our sensors differ.	The Sensor Twin learns each one from signals it already hears; the clock network separates sensor and emitter clocks given one reference.
What is the maturity?	TRL 2 today for the innovations (front-end TRL 4); targets per element in §10.
Can it run on our isolated network?	Yes: offline installer, pinned dependencies, SBOM, zero-network test in acceptance.
Blind LDPC?	Catalogue identification by syndrome; short-code recovery (n ≤ 1,024) as research mode; long-code blind recovery is not claimed.

Question	Answer
Encrypted payloads?	Recipe and frame structure recovered; payload flagged high-entropy; decryption out of scope.

13. References (verified 26 Sep 2026)

Research on blind identification, sensing and learning

#	Reference	Link
1	Singh et al., 'Blind Identification of Channel Codes: A Subspace-Coding Approach', arXiv 2601.15903, 2026	https://arxiv.org/abs/2601.15903
2	'Open-set blind recognition of non-binary LDPC codes', Digital Communications and Networks, 2026	https://doi.org/10.1016/j.dcan.2025.10.004
3	Ding et al., 'Joint Blind Identification of Cyclic Codes and Synchronization Scramblers From Synchronous Scrambling Sequences of Cyclic Codes Based on Soft Decision Information', IET Communications, 2026	https://doi.org/10.1049/cmu2.70192
4	Tixier, 'Blind identification of an unknown interleaved convolutional code', arXiv 1501.03715, 2015	https://arxiv.org/abs/1501.03715
5	'Blind Recognition of Reed-Solomon Codes Based on Galois Field Fourier Transform and Reliability Verification', IEEE Communications Letters, 2023	https://doi.org/10.1109/LCOMM.2023.3285607
6	Wei et al., 'Blind Recognition of Frame Synchronization Based on Deep Learning', Sensors, 2024	https://doi.org/10.3390/s24206767
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Hanley and Lippman-Hand, 'If Nothing Goes Wrong, Is Everything All Right?', JAMA, 1983	10.1001/jama.1983.03330370053031	C1, §9

Standards, programme references and tools

Item	Use
NASA NPR 7123.1 (Appendix E, TRL definitions); NASA Systems Engineering Handbook SP-2016-6105 Rev 2 (verification methods)	§4, §10
ITU-R F.1487 (HF channel simulation); ARINC 635 (HFDL); STANAG 4285; ITU-R M.1371 (AIS); ITU-R M.540 (NAVTEX); ETSI EN 302 307 (DVB-S2); CCSDS 131.0-B (TM coding)	§7
WPC G.S.R. 347(E), 2022 (433.05-434.79 MHz exemption); Indian Wireless Telegraphy Act, 1933	§7
SigMF; GNU Radio 3.10 (gr-digital, gr-fec); SatDump; gr-satellites; dumphfdl; fldigi; dump1090; AIS-catcher; KiwiSDR and kiwirecorder; SatNOGS network API	§5, §7, §8
Signal facts cross-checked against: Signal Identification Wiki (LRPT, STANAG 4285, HFDL); AMSAT and AMSAT-UK (FUNcube-1); AMSAT-DL and BATC (QO-100); Analog Devices wiki (ADALM-Pluto); RTL-SDR Blog V4 datasheet; kiwisdr.com	§7

Open items that paper cannot close: measured KPIs (after the build); SIH 2026 dates and PS clarifications (SPOC); equipment procurement; institute approval for the rig; live status of FUNcube-family satellites at the time of recording.